

SWADESH RANJAN DASH

FORWARD DEPLOYED ENGINEER / ARCHITECT • SOFTWARE ARCHITECT • AI-ASSISTED ENGINEERING

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EXECUTIVE PROFILE

Hands-on Software Architect and technical leader with 11+ years of experience designing, building, debugging and delivering scalable software systems across customer-facing and product engineering environments. Strong background in Python, distributed systems, microservices, APIs, concurrency, cloud architecture and performance engineering. Experienced in moving from customer requirements and technical discovery to architecture, hands-on implementation, debugging, validation and delivery. Recent hands-on adoption of Generative AI and Agentic AI for engineering workflows, including RAG, embeddings, vector search, LangChain/LangGraph, debugging, root-cause analysis, unit-test generation, change/impact analysis, Python 2 to Python 3 modernization and performance validation.

CORE EXPERTISE

ARCHITECTURE	ENGINEERING	AI / GENAI	LEADERSHIP
Solution Architecture - Distributed Systems - Microservices - AWS / Cloud	Python - Concurrency & Sockets - Debugging & RCA - Performance Engineering	Generative AI - Agentic AI - RAG / Embeddings / Vector Search - LangChain / LangGraph	Technical Leadership - Stakeholder Management - Team Mentoring - Customer Solutioning

TECHNICAL TOOLKIT

Languages & Frameworks: Python • Django • Flask • FastAPI • PHP • Node.js • JavaScript

Architecture: Solution Architecture • Microservices • REST APIs • Distributed Systems • Design Patterns • SOLID

AI / GenAI: Generative AI • Agentic AI • RAG • Embeddings • Vector Search • LangChain • LangGraph • AI-assisted Engineering

Data: Pandas • NumPy • MySQL • MS SQL • MongoDB • SQLite • JSON • XML • CSV

Cloud / DevOps: AWS • EC2 • S3 • Lambda • ELB • Docker • Kubernetes • Nginx • Gunicorn • RHEL • CI/CD

Security: JWT • HMAC • SSL/TLS

Engineering: Multithreading • Multiprocessing • Coroutines • Socket Programming • Performance Optimization • MPS

AI & ENGINEERING INNOVATION

- Created reusable AI-assisted engineering skills for issue investigation, code understanding, debugging and root-cause analysis.
- Worked on an Agentic AI-based Python 2 to Python 3 migration workflow to analyze legacy code, identify compatibility issues, support migration changes and accelerate validation.
- Developed workflows supporting unit-test generation and change/impact analysis to identify affected components before implementation and validation.
- Created a performance-testing skill that executes test scenarios and calculates Messages Per Second (MPS) for measurable system benchmarking.
- Applied RAG, embeddings, vector search, LangChain and LangGraph concepts within practical engineering workflows, using AI as an accelerator with deterministic validation and human review.

PROFESSIONAL EXPERIENCE

WIPRO LTD | Mar 2025 – Present

Technical Lead | Client: Lead Application / Software Architect

- Led technical discovery during the RFP phase, translating complex customer requirements into solution architecture and implementation direction.
- Designed and presented scalable architecture blueprints covering API integrations, multi-threaded components, data flows and reliability considerations.
- Rapidly gained deep understanding of a complex product involving Python concurrency, Unix services, daemon processes, coroutines and socket-based communication.
- Moved from architecture into hands-on engineering, resolving critical issues involving synchronization, deadlocks, race conditions and socket connection failures.
- Partnered with senior engineers and product owners to investigate issues, refine features and integrate changes across core modules, APIs and Unix backend services.
- Created reusable AI-assisted engineering skills for debugging, root-cause analysis, code understanding, unit-test generation and change/impact analysis.
- Created a performance-testing skill to execute scenarios and calculate MPS (Messages Per Second), enabling measurable performance validation.
- Worked on an Agentic AI-based Python 2 → Python 3 migration workflow, using AI-driven code analysis, dependency identification, migration support and validation to accelerate legacy modernization.
 - Promoted practical GenAI adoption within engineering workflows to improve productivity and accelerate complex problem resolution.
- Mentored engineers on concurrency, debugging, coroutines, system behavior and end-to-end product flow; recognized by the client for ownership and rapid issue resolution.

Technology: Python • Multithreading • Multiprocessing • Coroutines • Sockets • Unix • APIs • Distributed Systems • GenAI • Agentic AI • RAG • Embeddings • Vector Search • LangChain • LangGraph

CAPGEMINI TECHNOLOGY SERVICE INDIA LTD | Sep 2021 – Mar 2025

Technical Manager | Client: Lead Application / Software Architect

- Led a team of 15 developers across end-to-end delivery, mentoring, performance management and technical execution.
- Translated stakeholder requirements into software features, technical designs and scalable solution approaches across multiple projects.
- Designed and developed secure applications using Python, Flask and Django; deployed services with Gunicorn/Nginx on RHEL.
- Worked with microservices and AWS services including Lambda, S3 and EC2 to improve scalability, resource efficiency and reliability.
- Led integration of MS SQL/MySQL databases, REST APIs and secure interfaces using JWT/HMAC; drove design and code reviews using SOLID principles and design patterns.
- Supported Agile delivery through JIRA, Git/SVN/TFS and CI/CD; mentored developers on architecture, AWS best practices and serverless technologies.

Technology: Python • Flask • Django • AWS Lambda/S3/EC2 • Microservices • REST • JWT/HMAC • MySQL/MS SQL • CI/CD

EKINOPS NETWORKS INDIA PRIVATE LIMITED | Nov 2019 – Sep 2021

R&D Engineer (Lead)

- Led technical teams, distributed work and resolved technical roadblocks while delivering and maintaining web applications.
- Developed and maintained systems using Python, PHP, Node.js, MySQL, message queues, MQTT, WebSocket, Docker, AWS and REST/SOAP APIs.
- Provided technical guidance, mentoring and proactive production support with focus on performance, scalability and reliability.

Technology: Python • PHP • Node.js • MySQL • MQTT • Message Queue • WebSocket • Docker • AWS

SASKEN TECHNOLOGY PVT LTD | Sep 2018 – Oct 2019

Sr Engineer – Application Software Design & Development

- Developed web applications using Python/Django, PHP, Node.js, SQLite, messaging, WebSocket and REST/SOAP APIs.
- Supported CI/CD integration, technical task analysis, stakeholder coordination and mentoring of new joiners.

Technology: Python • Django • Node.js • SQLite • Messaging • WebSocket • REST/SOAP • CI/CD

PROVAB TECHNOSOFT | Mar 2017 – Jul 2018

Sr Software Developer

- Developed and maintained web applications using Python/Django, PHP, MySQL/SQLite and REST/SOAP APIs.
- Handled client requirements, bug fixing, technical analysis, deployments and DevOps activities.

Technology: Python • Django • PHP • MySQL • SQLite • REST/SOAP • DevOps

SKILLEDANSWERS INFO SOLUTION PVT LTD | May 2015 – Mar 2017

Sr Software Developer

- Developed web applications using Core PHP, Python, JavaScript and jQuery and integrated REST/SOAP web services.
- Worked directly with project management and clients to capture functional requirements and resolve assigned defects.

Technology: Python • PHP • JavaScript • jQuery • REST/SOAP

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate — Amazon Web Services
- SAFe® 6 Agilist — Scaled Agile, Inc.
- SAFe® 6 Scrum Master — Scaled Agile, Inc.
- Software Architecture — Coursera

EDUCATION

B.Tech — Electronics & Telecommunication Engineering | Ajay Binay Institute of Technology (BPUT), Cuttack, Odisha | 2013